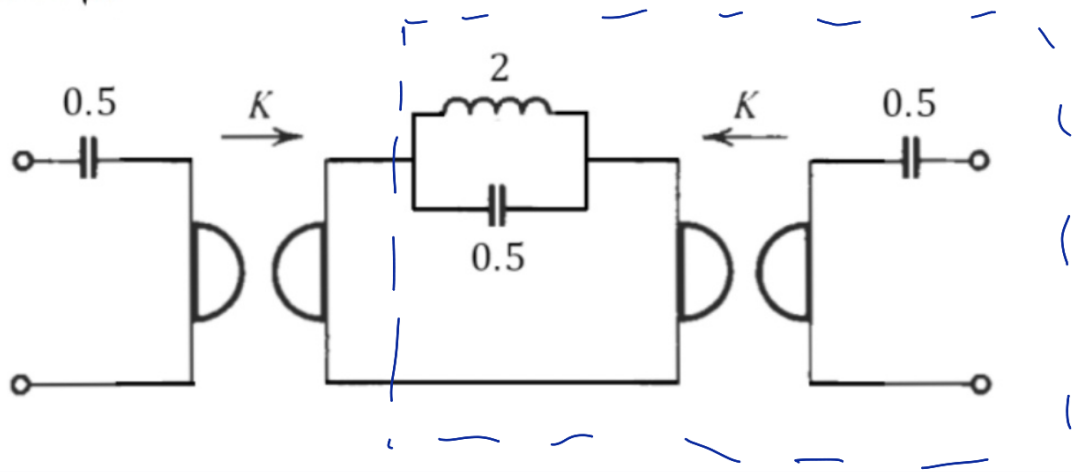


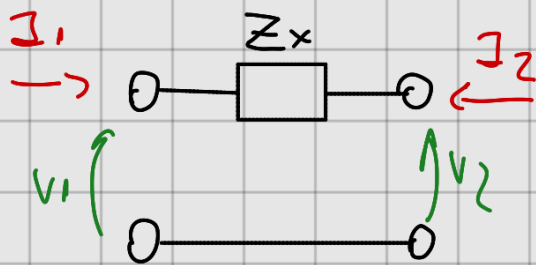
Ejercicio #6

Obtenga un circuito lattice equivalente al siguiente cuadripolo. Utilice la constante del girador $K = \sqrt{2}$



Trabajo con parámetros T.

comienzo con las siguientes definiciones,



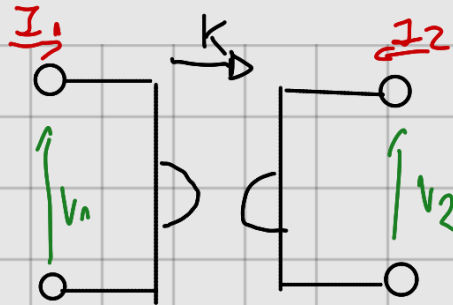
$$T_{xA} = \frac{V_1}{V_2} \Big|_{I_2=0} = 1$$

$$T_{xB} = \frac{V_1}{-I_2} \Big|_{V_2=0} = Z_x$$

$$T_{xC} = \frac{I_1}{V_2} \Big|_{I_2=0} = 0$$

$$T_x = \begin{pmatrix} 1 & Z_x \\ 0 & 1 \end{pmatrix}$$

$$T_{xD} = \frac{I_1}{-I_2} \Big|_{V_2=0} = 1$$



$$v_2 = I_1 \cdot k$$

$$v_1 = -I_2 \cdot k$$

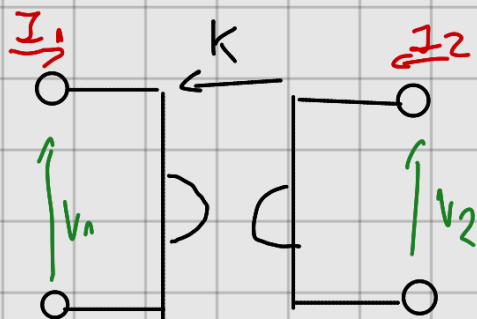
$$T_{G_A} = \left. \frac{v_1}{v_2} \right|_{I_2=0} = 0$$

$$T_{G_B} = \left. \frac{v_1}{-I_2} \right|_{v_2=0} = k$$

$$T_{G_C} = \left. \frac{I_1}{v_2} \right|_{I_2=0} = \frac{1}{k}$$

$$T_{G_D} = \left. \frac{I_1}{-I_2} \right|_{v_2=0} = 0$$

$$T_G = \begin{pmatrix} 0 & k \\ \frac{1}{k} & 0 \end{pmatrix}$$



$$v_2 = -I_1 \cdot k$$

$$v_1 = I_2 \cdot k$$

$$T_{G_A} = \left. \frac{v_1}{v_2} \right|_{I_2=0} = 0$$

$$T_{G_B} = \left. \frac{v_1}{-I_2} \right|_{v_2=0} = -k$$

$$T_{G_C} = \left. \frac{I_1}{v_2} \right|_{I_2=0} = -\frac{1}{k}$$

$$T_{G_D} = \left. \frac{I_1}{-I_2} \right|_{v_2=0} = 0$$

$$T_G = \begin{pmatrix} 0 & -k \\ -\frac{1}{k} & 0 \end{pmatrix}$$

En base a estas "secciones" T, obtengo el

T_{TOTAL} del diagrama

$$T_T = T_x \Big|_{z_x = \frac{1}{sC}} \cdot T_G \cdot T_x \Big|_{z_x = \frac{1}{sC} // sL} \cdot T_{G1} \cdot T_x \Big|_{z_x = \frac{1}{sC}}$$

$$T_T = \begin{pmatrix} 1 & \frac{1}{sC} \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & k \\ \frac{1}{k} & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & \frac{S/C}{s^2 + 1/LC} \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & -k \\ -\frac{1}{k} & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & \frac{1}{sC} \\ 0 & 1 \end{pmatrix}$$

$$T_T = \begin{pmatrix} \frac{1}{sC} \cdot \frac{1}{k} & k \\ \frac{1}{k} & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & \frac{S/C}{s^2 + 1/LC} \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & -k \\ -\frac{1}{k} & -\frac{1}{k} \cdot \frac{1}{sC} \end{pmatrix}$$

$$T_T = \begin{pmatrix} \frac{1}{sC} \cdot \frac{1}{k} & \frac{1}{sC} \cdot \frac{S/C}{s^2 + 1/LC} + k \\ \frac{1}{k} & \frac{1}{k} \cdot \frac{S/C}{s^2 + 1/LC} \end{pmatrix} \cdot \dots$$

$$T_T = \begin{pmatrix} -\frac{1}{k} \cdot \frac{1}{sC} \cdot \frac{S/C}{s^2 + 1/LC} - 1 & -\frac{1}{sC} - \frac{1}{k} \cdot \frac{1}{s^2 C^2} \cdot \frac{S/C}{s^2 + 1/LC} \cdot \frac{1}{sC} \\ -\frac{1}{k^2} \cdot \frac{S/C}{s^2 + 1/LC} & -\frac{1}{k^2} - \frac{1}{k^2} \cdot \frac{1}{sC} \cdot \frac{S/C}{s^2 + 1/LC} \end{pmatrix}$$

$$T_T = \begin{pmatrix} 2 \cdot \frac{1}{s \cdot (s^2 + 1)} + 1 & 4 \cdot \frac{1}{s} + 4 \cdot \frac{1}{(s^2 + 1) \cdot s} \\ \frac{1}{s^2 + 1} & 1 + 2 \cdot \frac{1}{s^2 + 1} \end{pmatrix} \cdot (-1)$$